

A Regularized Objective for Robust Image Classification

Abstract

We propose a regularized training objective and evaluate it on standard image classification benchmarks. Full results appear in Section 4.

Section 2

[illegible]

Section 3

[illegible]

Methods

We evaluate our approach on five datasets spanning natural images, digits and apparel classification. All models share the same training schedule.

Training runs for 50 epochs with a batch size of 128 on a single GPU.

Section 5

[illegible]

Section 6

[illegible]

Results

Our method improves accuracy by 12% over the baseline across all evaluated datasets. Gains are consistent and largest on SVHN.

Table 2 reports per-dataset accuracy for the baseline and our method.

Section 8

[illegible]

Discussion

The approach is also more robust to input noise, which we attribute to the regularizing effect of the proposed objective.

Section 10

[illegible]

Section 11

[illegible]

Section 12

[illegible]